Ex3b: Model Planning and Building - Regression

1. Learning Objectives

- To understand the concept of regression in machine learning.
- To load and explore the Advertising dataset.
- To identify the relationship between TV, Radio, Newspaper advertising and Sales.
- To build different regression models for predicting Sales.
- To compare Linear Regression, Polynomial Regression, and SVR models.
- To evaluate model performance using MAE, MSE, RMSE, and R^2 Score.
- To identify the best-performing regression model.
- To use model results for making business decisions about advertising budget.

2. Methods Used / Python Libraries Involved

Library / Method	Purpose	Reason to Use
Pandas	Load and manipulate the dataset	Easy handling of tabular data
NumPy	Numerical calculations	Used for calculating RMSE and array operations
train_test_split	Split data into training and testing sets	To test the model on unseen data
LinearRegression	Build Linear Regression model	Simple and effective regression method
PolynomialFeatures	Convert features into polynomial features	Helps model non-linear relationships

3. Program Code

```
import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns from sklearn.model_selection import train_test_split from sklearn.linear_model import LinearRegression from sklearn.preprocessing import PolynomialFeatures, StandardScaler from sklearn.pipeline import make_pipeline from sklearn.svm import SVR from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
df = pd.read_csv("Advertising.csv") df =
  df.drop(columns=["Unnamed: 0"]) df.columns =
  ["TV", "Radio", "Newspaper", "Sales"]

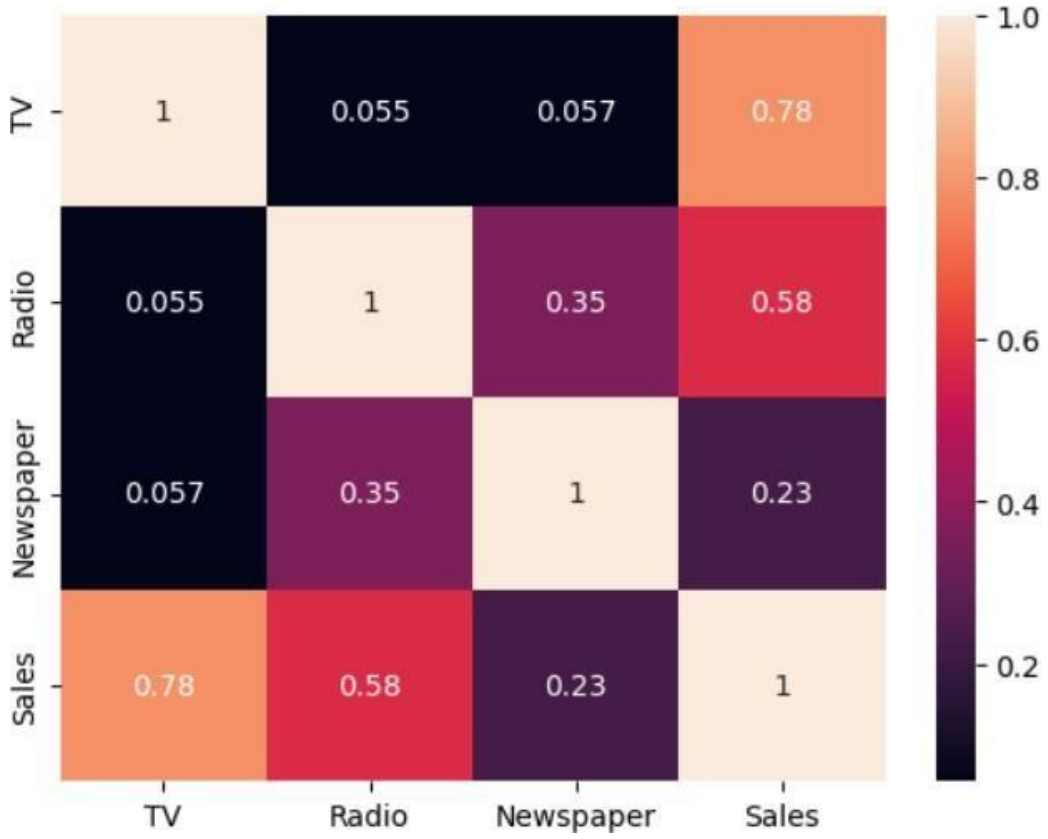
  X = df[["TV", "Radio", "Newspaper"]]
  y = df["Sales"]
print(df.head())
  print(df.shape)
  print(df.isnull().sum()
  )
sns.pairplot(df)
  plt.show()
sns.heatmap(df.corr(), annot=True)
  plt.show()

  X_train, X_test, y_train, y_test = train_test_split(
      X, y, test_size=0.2, random_state=42
  ) models
  = {
      "Linear": LinearRegression(),
      "Polynomial": make_pipeline(
          PolynomialFeatures(2),
          LinearRegression()
      ),
      "SVR": make_pipeline(
          StandardScaler(),
          SVR(kernel="rbf")
      ) }
  results =
  [] for
  name, model
  in
  models.item
  s():
  model.fit(X
  _train,
  y_train)
  pred =
  model.predi
  ct(X_test)
  mae =
  mean_absolu
  te_error(y_
  test, pred)
  mse =
  mean_square
  d_error(y_t
  est, pred)
  rmse =
  np.sqrt(mse
```

```
) r2 =
r2_score(y_
test, pred)
results.app
end([name,
mae, mse,
rmse, r2])
results = pd.DataFrame( results, columns=["Model",
    "MAE", "MSE", "RMSE", "R2"]
)       print("\nModel       Comparison:")
print(results)
results.set_index("Model").plot(kind="bar")
plt.title("Regression  Model  Comparison")
plt.ylabel("Score")   plt.show()   best   =
results.loc[results["RMSE"].idxmin()]
print("\nBest    Model:",    best["Model"])
print("RMSE:",    round(best["RMSE"],    4))
print("R2 Score:", round(best["R2"], 4))
```

4. Output

```
TV Radio Newspaper Sales
0 230.1 37.8        69.2 22.1
1  44.5 39.3        45.1 10.4
2  17.2 45.9        69.3      9.3
3 151.5 41.3        58.5 18.5
4 180.8 10.8        58.4 12.9
(200, 4)
TV          0
Radio       0
Newspaper      0
Sales   0
dtype: int64
```



Model Comparison:

	Model	MAE	MSE	RMSE	R2
0	Linear	1.460757	3.174097	1.781600	0.899438
1	Polynomial	0.526179	0.412910	0.642581	0.986918
2	SVR	1.162139	2.862109	1.691777	0.909322

5. Learning Outcome

- I learned how to load and explore a real-world dataset using Pandas.
- I learned how to perform Exploratory Data Analysis (EDA) using pair plots and correlation heatmaps.
- I understood how TV, Radio, and Newspaper advertising affect Sales.
- I learned how to split a dataset into training and testing data.
- I learned to build Linear Regression, Polynomial Regression, and SVR models.
- I learned how to evaluate regression models using MAE, MSE, RMSE, and R^2 Score.